

Find $P[X, Y \in A]$ where A is the region $(x, y) | 0 < x < 1/2; 1 < y < 2$

$$P[X, Y \in A] = P(0 < x < 1/2; 1 < y < 2)$$

$$\begin{aligned} &= \int_1^2 \int_0^{1/2} \frac{3}{5} x(y+x) dx dy \\ &= \int_1^2 \left[\frac{3yx^2}{10} + \frac{3x^3}{15} \right] dy = \int_1^2 \left(\frac{3y}{40} + \frac{1}{40} \right) dy = \left(\frac{3y^2}{80} + \frac{1}{40} \right)_1^2 = \frac{11}{80} \end{aligned}$$

If X and Y are continuous random variables, the cumulative density or distribution function is given by

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(s, t) ds dt;$$

for $-\infty < x < \infty; -\infty < y$

where, $f(s, t)$ is the value of the joint probability distribution of X and Y at (s, t) and is called the joint distribution or the **joint distribution function of X and Y** .

We assume that the joint distribution is continuous everywhere and partially differentiable with respect to each variable, for all but a finite set of values of the two random variables and is analogous of the relationship $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$;

whenever this partial derivative exists. The joint distribution of two continuous variables determines their joint density at all point (x, y) , where the joint density is continuous.

Example: $f(x, y) = \begin{cases} (x+y) & \text{for } 0 < x < 1; 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$

Region I: For $0 < x < 1; 0 < y < 1$; we get

$$F(x, y) = \int_0^y \int_0^x (x+y) dx dy = \frac{1}{2} xy(x+y);$$

Region II: For $x > 1; 0 < y < 1$, we get

$$F(x, y) = \int_0^y \int_0^1 (x+y) dx dy = \frac{1}{2} y(1+y)$$

Region III: For $0 < x < 1, y > 1$; we get

$$F(x, y) = \int_0^1 \int_0^x (x+y) dx dy = \frac{1}{2} x(1+x)$$

Region IV: For $x > 1, y > 1$; we get

$$F(x, y) = \int_0^1 \int_0^1 (x+y) dx dy = 1$$

Since the joint distribution function is continuous everywhere, the boundaries between any two of these regions can be included in either one and we can write

$$F(x, y) = \begin{cases} 0, & 0 \leq x \leq 0, y \leq 0 \\ \frac{1}{2}xy(x+y); & 0 < x < 1, 0 < y < 1 \\ \frac{1}{2}y(y+1); & x \geq 1, 0 < y < 1 \\ x(x+1); & 0 < x < 1, y \geq 1 \\ 1; & x \geq 1, y \geq 1 \end{cases}$$

27.3 MARGINAL DISTRIBUTION

Now, there would be cases where we might need to remove the influence of an event. This might happen if some events are irrelevant for some experiment or we could alternatively be only interested in a subset of the events. Therefore, we resort to ‘marginalisation’ which refers to the process of removing/omitting the influence of one or more events.

If X and Y are discrete random variables and $f(x, y)$ is the value of their joint probability distribution at (x, y) , then the function given by

$$g(x) = \sum_y f(x, y) \text{ for each value of } x \text{ within the range of } X \text{ is called the}$$

marginal distribution of X and correspondingly the function given by $h(y) = \sum_x f(x, y)$ or each value of y within the range of Y is called the **marginal**

distribution of Y .

If X and Y are the **continuous random variables** and $f(x, y)$ is the value of their joint probability density at (x, y) then the function given by

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy; \text{ for } -\infty < x < \infty \text{ is called the } \mathbf{marginal \text{ density of } X} \text{ and the}$$

corresponding function given by $h(y) = \int_{-\infty}^{\infty} f(x, y) dx; \text{ for } -\infty < y < \infty \text{ is called the}$

marginal density of Y

Example: We continue the example discussed for joint distribution and calculate marginal distribution (Refer Table 27.3& 27.4):

Table 27.3: Marginal Distribution of X

		X			
		2	3	4	5
y	0	0	0.14	0	0
	1	0.35	0.3	0.05	0.05
	2	0	0	0.11	0
	$\sum_y f(x,y)$	0.35	0.44	0.16	0.05

Table 27.4 Marginal Distribution of Y

y		X				
		2	3	4	5	$\sum_x f(x,y)$
	0	0	0.14	0	0	0.14
	1	0.35	0.3	0.05	0.05	0.75
	2	0	0	0.11	0	0.11

Check Your Progress 1

- 1) Let the joint probability density function for (X,Y) be

$$f(x,y) = \frac{x+y}{4}, x > 0, y > 0, 3x + y < 3,$$

zero otherwise.

Find the marginal probability density function of i) X, $f_X(x)$ ii) Y, $f_Y(y)$

.....

- 2) Given the joint probability density

$$f(x,y) = \begin{cases} \frac{2}{3}(x+2y); & \text{for } 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the marginal densities of X and Y

.....

27.4 THREE IMPORTANT PROBABILITY DISTRIBUTIONS

In real life, we come across various situations that can be modelled using probability distributions. This helps in understanding the probability of a certain occurrence. Therefore, based on distinct characteristics of different situations under consideration, we resort to the different probability distribution that best suits the characteristics.

The following section will deal with the three most common probability distributions. Here, we start off with the discussion of Binomial distribution leading to Poisson and then lastly Normal distribution.

27.4.1 Binomial Distribution

Repeated trials play a very important role in probability and statistics, especially when the number of trials is fixed and the probability of success (or failure) in each of the trial is independent.

The theory, which we discuss in this section, has many applications; for example, it applies to events like the probability of getting 5 heads in 12 flips of a coin or the probability that 3 persons out of 10 having a tropical disease will recover. To apply binomial distribution in these cases, the probability (of getting head in each flip and recovering from the tropical disease for each person) should exactly be the same. More importantly, each of the coin tossed and chance of recovering of each patient should be independent.

Let x be a discrete random variable following a binomial distribution. The sampling distribution of x is the probability mass function (p.m.f) and is defined by $f(x) = {}_n C_x P^x Q^{n-x}; x = 0, 1, 2, \dots, n; (P + Q) = 1$; where P and Q are positive fractions.

Suppose that we have a series of independent trials, in each of which the probability of occurrence of an event is fixed and constant (say P). Let us denote the occurrence of the event as success and its complement that is the nonoccurrence of the event as failure. Then the probability that an event occurs exactly r times in n trials is ${}_n C_r P^r Q^{n-r}$; where $Q = 1 - P$ and r may assume values $= 0, 1, 2, \dots, n$. Thus, in a series of n independent trials, if the probability of success in each trial is a constant (say P) and the probability of failure is Q , then the probability of x success (and obviously $n - x$ failure) is given by the Binomial distribution. The two constants n and P appearing in the p.m.f $f(x)$ are known as the parameters of Binomial distribution. If the values of n and P are known the distribution is completely known. It should be noted that the binomial distribution holds under following conditions:

- 1) The outcomes of any trial can be classified under only two categories (say success and failure)
- 2) The probability of success of each trials remain constant and does not change from one trial to another
- 3) The trials are independent, so that the probability of success in any trial remains unaffected by the results of other trials.

Example: Find the probability of getting 7 heads and 5 tails in 12 tosses of an unbiased coin.

Substituting $x = 7$, $n = 12$, $p = \frac{1}{2}$ in the formula of binomial distribution, we get

the desired probability of getting 7 heads and 5 tails.

$$B(7;12,1/2) = {}^{12}C_7 \cdot 1/2^7 \cdot (1 - 1/2)^5 = {}^{12}C_7 \cdot (1/2)^{12}$$

There are a few important properties of the binomial distribution. While discussing these, we retain the notations used earlier in this unit.

- Binomial distribution is a discrete probability distribution where the random variable assumes a finite number of values (0, 1, 2, 3, ..., n). The distribution is specified by two parameters n and P.
- Mean = n P; Variance = n P Q (where n is the number of repeated trials and P and Q are the probability of success and failure).
- Binomial distribution may have either one or two modes. When (n+1)P is not an integer, mode is the largest integer contained therein. However, when (n+1)p is an integer, there are two modes given by (n + 1)P and {(n + 1)P - 1}.
- Skewness of binomial distribution is given by $\frac{Q-P}{\sqrt{nPQ}}$; Kurtosis = $\frac{1-6PQ}{nPQ}$.

When P = Q = 1/2 the distribution becomes symmetric for all values of n.

- If X and Y are two random variables, where X follows binomial distribution with parameters (n₁, p) and Y follows binomial distribution with parameters (n₂, p), then the random variable (X+Y) also follows binomial distribution with parameters (n₁+n₂, p).
- The moment generating function of a binomial distribution is given by

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \sum_{i=0}^n e^{tx} f(x) = \sum_{i=0}^n e^{tx} \cdot {}^nC_x \cdot p^x (1-p)^{n-x} \\ &= \sum_{i=0}^n {}^nC_x (pe^t)^x (1-p)^{n-x} \end{aligned}$$

The summation is easily recognized as the binomial expansion of $\{p \cdot e^t + (1-p)\}^n = \{1 + p(e^t - 1)\}^n$

We can derive the mean and variance of the binomial distribution using the moment generating function.

If we differentiate $M_X(t)$ with respect to t twice, we get:

$$M'_X(t) = npe^t [1 + p(e^t - 1)]^{n-1}$$

$$M''_X(t) = npe^t [(1 + p(e^t - 1))^{n-1} + n(n-1)e^{2t}p^2 [1 + p(e^t - 1)]^{n-2}]$$

$$M''_X(t) = npe^t (1-p + npe^t) [1 + p(e^t - 1)]^{n-2}$$

And substituting $t = 0$, we get $\mu'_1 = np$ and $\mu'_2 = np(1-p + np)$. Thus, $\mu = np$ and

$$\sigma^2 = \mu'_2 - (\mu'_1)^2 = np(1-p + np) - (np)^2 = np \cdot (1-p)$$

We can verify that the mean and the variance of binomial distribution are $\mu = np$ and $\sigma^2 = np(1-p)$, when n is the number of trials and p is probability of success and $(1-p) = q$ is the probability of failure as follows:

$$\begin{aligned}\mu &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x}\end{aligned}$$

We have omitted the term $x = 0$ and cancelled the x against the first factor of $x! = x(x-1)!$ in the denominator of $\binom{n}{x}$. Then factoring out the factor $n! = n(n-1)!$ and one factor p we get

$$\mu = np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x}$$

Letting $y = x - 1$ and $m = n - 1$, this becomes

$$\mu = np \sum_{y=0}^m \binom{m}{y} p^y (1-p)^{m-y} = np$$

The last summation is the sum of all the values of a binomial distribution with the parameters n and p and hence equal to 1.

To find the expression of μ'_2 and σ^2 , let us make use of the fact that $EX^2 = E[X(X-1)] + E(X)$ and first evaluate $E[X(X-1)]$. Duplicating for all practical purposes the step used before, we get :

$$\begin{aligned}E[X(X-1)] &= \sum_{x=0}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=2}^n \frac{n!}{(x-2)!(n-x)!} p^x (1-p)^{n-x} \\ &= n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} (1-p)^{n-x}\end{aligned}$$

Letting $y = x - 2$ and $m = n - 2$, this becomes

$$\begin{aligned}E[x(x-1)] &= n(n-1)p^2 \sum_{y=0}^m \binom{m}{y} p^y (1-p)^{m-y} \\ &= n(n-1)\sigma^2\end{aligned}$$

Therefore $\mu'_2 = E[X(X-1)] + E(X) = n(n-1)\sigma^2 + np$

$$\text{Finally, } \sigma^2 = \mu'_2 - \mu^2 = n(n-1)\sigma^2 + np + n^2p^2 = np(1-p)$$

Example: Three coins are tossed. Find the probability of (i) 0 head, 1 head, 2 heads, 3 heads (ii) more than one head (iii) at least one head.

Ans. Let the random variable x is the number of heads obtained in a toss of three

coins. If we denote the occurrence of head as success and the non-occurrence as failure and the coins are assumed to be unbiased then let

P = Probability of success in a single trial

= Probability of head in a single trial

$$= \frac{1}{2}$$

$$Q = 1 - P = 1 - \frac{1}{2} = \frac{1}{2}$$

n = Number of independent trials.

Since the value of P is constant and for each coin the trials are independent, the variable x follows a binomial distribution with parameters $n = 3$ and $p = \frac{1}{2}$. Thus, probability of x success is $f(x) = {}_n C_x P^x Q^{n-x} = {}_3 C_x (1/2)^x (1/2)^{3-x} = {}_3 C_x (1/2)^3$

1) Putting the values of $x = 0, 1, 2, 3$ respectively

$$f(0) = \text{Probability of 0 head} = {}_3 C_0 (1/2)^3 = 1 \times 1/8 = 1/8$$

$$f(1) = \text{Probability of 1 head} = {}_3 C_1 (1/2)^3 = 3 \times 1/8 = 3/8$$

$$f(2) = \text{Probability of 2 heads} = {}_3 C_2 (1/2)^3 = 3 \times 1/8 = 3/8$$

$$f(3) = \text{Probability of 3 heads} = {}_3 C_3 (1/2)^3 = 1 \times 1/8 = 1/8$$

2) Prob. of more than one head = $f(2) + f(3) = 3/8 + 1/8 = 4/8 = \frac{1}{2}$

3) Prob. of at least one head = $1 - \text{Prob}(0 \text{ head}) = 1 - f(0) = 1 - 1/8 = 7/8$

Proposition: If x_1 follows a Binomial distribution with parameters (m_1, P) and x_2 follows a Binomial distribution with parameters (m_2, P) , and x_1 and x_2 are statistically independent, then $(x_1 + x_2)$ also follows Binomial distribution with parameters $(m_1 + m_2, P)$.

The result can be extended to several independent variables with a common P .

Proof: Suppose x_1 and x_2 are distributed independently in the binomial form with parameters m_1, P and m_2, P respectively. Then the distribution of the sum $(x_1 + x_2)$ can take values $0, 1, 2, \dots, m_1 + m_2$.

$$\begin{aligned} \text{Also, } P[x_1, x_2 = k] &= \sum_{k_1=0}^k P[x_1 = k_1] P[x_2 = k - k_1] \\ &= \sum_{k_1=0}^k \binom{m_1}{k_1} \binom{m_2}{k-k_1} p^{k_1} (1-p)^{m_1-k_1} p^{k-k_1} (1-p)^{m_2-k+k_1} \\ &= p^k (1-p)^{m_1+m_2-k} \sum_{k_1=0}^k \binom{m_1}{k_1} \binom{m_2}{k-k_1} \end{aligned}$$

Now, this sum is nothing but the sum of products of the coefficients of t^{k-1} in $(1+t)^{m_1}$ and t^{k-k_1} in $(1+t)^{m_2}$ for varying k_1 , the hence equals the coefficients of t^k in $(1+t)^{m_1+m_2}$, which is $\binom{m_1+m_2}{k}$

$$\text{Thus, } P[x_1 + x_2 = k] = \binom{m_1 + m_2}{k} p^k (1-p)^{m_1 + m_2 - k}$$

This shows that $x_1 + x_2$ is itself binomially distributed with parameters $m_1 + m_2$ and p . We also get from the general result that if $x_1, x_2, \dots, x_n$ are independently distributed binomial variables with parameters $m_1, p; m_2, p; \dots, m_n, p$; then the sum $x_1 + x_2 + \dots + x_n$ is also a binomial variable with parameters $m_1 + m_2 + \dots + m_n$ and p .

This implies that if $x_1, x_2, \dots, x_n$ are a random sample from a binomial distribution with parameters of the statistic, $x_1 + x_2 + \dots + x_n$ is also binomial with parameters n, m and p .

27.4.2 Poisson Distribution

When the number of trials discussed in the case of binomial distribution is very large, the calculation of the probabilities with binomial distribution becomes very cumbersome. Suppose we want to know what is the probability that in a clinical test 200 out of 300 mice will survive, after being infected by a virus. There is 50% chance that each mouse can fight with the virus by producing antibodies. If we use binomial distribution, the required probability is given by ${}^{300}C_{200} \cdot \frac{1}{2}^{200} \cdot \frac{1}{2}^{100}$. In this section, we will discuss a probability distribution, which can be used to approximate binomial probabilities of this kind. Specifically, we will investigate the limiting form of the binomial distribution when $n \rightarrow \infty$ and $p \rightarrow 0$, while $n \cdot p$ remains constant. Let us suppose $\lambda = n \cdot p$, therefore $p = \lambda/n$.

We can also write the binomial distribution as

$$\begin{aligned} b(x, n, p) &= {}^nC_x (\lambda/n)^x (1 - (\lambda/n))^{n-x} \\ &= \frac{n(n-1)(n-2)(n-3)\dots(n-x+1)}{x!} \times (\lambda/n)^x (1 - \lambda/n)^{n-x} \\ &= 1 \cdot (1 - 1/n)(1 - 2/n)(1 - 3/n)\dots(1 - (x-1)/n) / x! \times \lambda^x \times [(1 - (\lambda/n))^{-n/\lambda}]^{-\lambda} \times (1 - (\lambda/n))^{n-x} \end{aligned}$$

Finally, if we have $n \rightarrow \infty$ while x and p are constants

$$\begin{aligned} (1 - 1/n)(1 - 2/n)(1 - 3/n)\dots(1 - (x-1)/n) &\rightarrow 1 \\ (1 - (\lambda/n))^{-x} &\rightarrow 1 \\ (1 - (\lambda/n))^{-n/\lambda} &\rightarrow e \end{aligned}$$

Therefore, the limiting form of the binomial distribution becomes $P(x, \lambda) = \lambda^x e^{-\lambda} / x!$ for $x = 0, 1, 2, 3, \dots$ which is the p.m.f of a poisson random variable

Thus, in the limit when $n \rightarrow \infty$, $p \rightarrow 0$ and $np = \lambda$ remains constant; the number of success is a random variable which follows a poisson distribution with only one parameter λ . This distribution is named after the French mathematician Simeon Denis Poisson.

Following random variables are classic examples of poisson distribution.

- 1) Number of accidents on a road crossing.
- 2) Number of defects per unit area of a sheet material.
- 3) Number of telephone calls received by a telephone attendant.
- 4) Number of suicides in a year in an area etc.

Poisson distribution is a discrete probability distribution and is defined by the

$$\text{p.m.f: } f(x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots, \infty;$$

where λ is positive and is known as the parameter of Poisson distribution and e is the mathematical constant which is equal to 2.718 approx.

There are few important properties of the poisson distribution.

- Poisson distribution is a discrete probability distribution, where the random variable assumes countably infinite number of values such as 0, 1, 2, 3, to ∞ . The distribution is completely specified if the parameter λ is known.
- Mean and variance of poisson distribution are the same, both being λ .
- Poisson distribution, like the binomial distribution, may have either one or two modes. When λ is not an integer, mode is the largest value contained in λ and when λ is an integer, there are two modes, λ and $(\lambda-1)$
- The distribution has skewness = $1/\sqrt{\lambda}$ and kurtosis = $1/\lambda$. Therefore, we can conclude that poisson distribution is positively skewed and leptokurtic.
- If X_1 and X_2 are two random variables following the poisson distribution with parameters λ_1 and λ_2 respectively, then the random variable $(X_1 + X_2)$ also follows poisson distribution with the parameter $(\lambda_1 + \lambda_2)$.
- As we have discussed earlier, the poisson distribution could be used as an approximation to binomial distribution when n is large but np is fixed.
- The moment generating function of the poisson distribution is given by

$$M_x(t) = E(e^{tX}) = e^{\lambda(e^t - 1)}$$

$$\begin{aligned} M_x(t) &= \sum_{i=0}^{\infty} e^{tx} \cdot f(x) \\ &= \sum_{i=0}^{\infty} e^{tx} \cdot \lambda^x e^{-\lambda} / x! \\ &= e^{-\lambda} \sum_{i=0}^{\infty} e^{tx} \cdot \lambda^x / x! \end{aligned}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} (\lambda e^t)^x / x!$$

In the above expression $\sum_{i=0}^{\infty} (\lambda e^t)^x / x!$ can be recognized as the Maclaurin's series of e^z where $z = \lambda e^t$.

Thus, the moment generating function of the Poisson distribution is

$$M_x(t) = e^{-\lambda} \cdot e^{\lambda e^t}$$

$$M_x''(t) = \lambda \cdot e^t \cdot e^{\lambda(e^t-1)} + \lambda^2 \cdot e^{2t} \cdot e^{\lambda(e^t-1)}$$

Therefore, $\mu_1' = M_x'(0) = \lambda$ and $\mu_2' = M_x''(0) = \lambda + \lambda^2$ and we get $\mu = \lambda$ and $\sigma^2 = \mu_2' - (\mu_1')^2 = \lambda + \lambda^2 - \lambda^2 = \lambda$

Example: Let X be a random variable following poisson distribution. If $P(X=1) = P(X=2)$, find $P(X=0 \text{ or } 1)$ and $E(X)$.

For poisson distribution, the p.m.f. is given by $P(x, \lambda) = \lambda^x e^{-\lambda} / x!$

Therefore, $P(X=1) = \lambda^1 e^{-\lambda} / 1! = \lambda e^{-\lambda}$

$P(X=2) = \lambda^2 e^{-\lambda} / 2! = \lambda^2 e^{-\lambda} / 2$

As $P(X=1) = P(X=2)$, from the equation $\lambda e^{-\lambda} = \lambda^2 e^{-\lambda} / 2$, we get $\lambda = 2$.

Therefore, $E(X) = \lambda = 2$ and

$P(X=0 \text{ or } 1) = P(X=0) + P(X=1) = 2^0 e^{-2} / 0! + 2^1 e^{-2} / 1! = 3 \cdot e^{-2}$.

Example: A random variable x follows a Poisson distribution with parameter 3. Find the probabilities that the variable assume the values: (i) 0, 1, 2, 3 (ii) less than 3 (iii) at least 2. Given that $e^{-3} = .0498$.

Ans. For the Poisson distribution with parameter $\lambda = 3$, the probability of x success is:

$$f(x) = \frac{e^{-3} 3^x}{x!}$$

i) Putting $x = 0, 1, 2, 3$ successively the required probabilities are:

$$f(0) = \frac{e^{-3} 3^0}{0!} = e^{-3} = .0498$$

$$f(1) = \frac{e^{-3} 3^1}{1!} = 3e^{-3} = 3(0.0498)$$

$$f(2) = \frac{e^{-3} 3^2}{2!} = \frac{9(0.0498)}{2}$$

$$f(3) = \frac{e^{-3} 3^3}{3!} = \frac{27(0.0498)}{6}$$

ii) Probability less than 3 success is the sum of the probabilities: $f(0) + f(1) + f(2)$